

MFFTD: A Multiscale Feature Fusion Transformer Detector for Electricity Theft Based on Semi-Supervised Learning

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Abstract—The wide deployment of advanced metering infrastructure (AMI) in power systems allows utility companies to automatically and accurately collect and process the time-series load profiles of households, but meanwhile incurs the severe electricity theft (ET) that some illegal residential users may manipulate their electricity consumptions to reduce their billings. Although, due to the powerful ability of modeling long-range dependencies in sequential data, transformer has been widely used for time-series modeling including ET detection (ETD), the significant weakpoint lies in that it only considers the attention weights between either points or prepatched subsequences (i.e., patches) of fixed size within the input sequence, which cannot fully characterize the relationships among multiscale temporal patches and lead to suboptimal detection performance. To address the above issue, based on self-supervised feature extraction and supervised fine-tuning, our work proposes a novel multiscale feature fusion transformer encoder (TE)-based ETD framework, MFFTD. Specifically, our work's contributions are following. First, in MFFTD, a hierarchical patching enhanced TE (HPTE) is explicitly designed, in which each layer patches the input sequence with variable patch size. Then, through hierarchically stacked multiple HPTE layers, the feature combining multiscale patches can be effectively extracted. Second, considering the constraint that only a small percentage of labeled theft samples are practically available, our work first pretrains the structure parameters of MFFTD through a self-supervised pretext task of forecasting the randomly masked segments in time series. Then, the small percentage of labeled anomalous samples is used to fine-tune the MFFTD model. Extensive experiments on multiple real datasets demonstrate our proposed MFFTD scheme outperforms the state-of-the-art (SOTA) transformer-based supervised and semi-supervised ETD methods.

Index Terms—Electricity theft detection (ETD), multiscale feature fusion, self-supervised learning, semi-supervised learning, transformer, variable patching.

I. INTRODUCTION

THE advanced metering infrastructure (AMI) has been widely deployed into modern power systems, which

provides two-way communications between customers and utility companies [1]. Through AMI, utility companies are able to gather sequentially temporal information about individual consumption, i.e., time series, and accordingly charge customers. Wide deployment of AMI is a milestone that denotes traditional power systems are evolved into smart grids, where pervasive control at all levels is a basic premise [2]. Although many benefits have been brought by AMI, it also introduces a number of security vulnerabilities. For example, adversaries can launch a variety of cyberphysical attacks against AMI devices and network components [3]. One of the main purposes of these adversaries is to manipulate their electricity consumptions such that their billings can be lowered down, so-called the behavior of electricity theft (ET) [4].

A. Research Motivation

Recently, based on the massive amount of data provided by AMI, deep neural network (DNN)-based ET detection (ETD) schemes have been widely developed [5], [6], [7], since DNNs can automatically extract features and have powerful functional approximation ability, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs). Technically, CNN-based detectors [8], [9] transform 1-D temporal electricity consumption data into 2-D matrix to extract spatial feature. Different from CNN-based detectors, long short-term memory (LSTM, a specific variant of RNN)-based ET detectors accept time series sequentially as input and inherently learn the temporal dependencies of electricity consumption data [10], [11]. However, their weakpoint lies in that: LSTM cannot compute in parallel and cannot efficiently learn a long-range temporal dependencies and CNN-based detectors mainly focus on extracting local features of the electricity consumption data. Although there existed work in literature that design two-level CNNs [12] or adopt ConvLSTM block [13] to attempt to simultaneously capture the long-range and local features, both the structures still have the problem of not sufficiently learning global information of the input temporal sequence. Recently, transformer-based detectors [14], [15] have been proposed to solve the aforementioned issues. Transformer is built on multihead attention mechanism and is regarded as suitable modeling time series including ETD task: It concurrently represents each input sequence element by considering its context (future–past), while multiple attention heads can incorporate different representation subspaces [16]. However, there are two limitations

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in the existing transformer-based ET detectors. The first is that patching with fixed size used in traditional transformer [17] cannot sufficiently capture multiscale feature. The multiscale feature fusion method, by combining information from different scales, can effectively improve the detection accuracy of ETD systems. In [12] and [18], CNNs are used to extract features at different time scales from electricity consumption data. However, the aforementioned CNN-based methods primarily focus on local features and are not effective at capturing global and temporal information in electricity consumption data.

The second is the so-called insufficiency of labeled anomalous samples. Since extensive data labeling is often prohibitively expensive or impractical: require the knowledge of domain expertise, therefore, it is imperative to achieve high detection performance using only a limited amount of labeled data or by leveraging the existing plethora of unlabeled data. Generally, pretraining and fine-tuning learning paradigm is well-suited to address this issue. In this paradigm, the model is first pretrained for a pretext task on a dataset containing large number of normal samples, and then fine-tuned with few anomalous samples for a specific downstream task, such as ETD. Following this paradigm, several TE-based architectures have been proposed, including time-series transformer (TST) [19] and Patch TST (PatchTST) [17]. Specifically, to extract the temporal relations in the time series, patching is usually used, in which the long temporal sequence is divided into several subsequences with a certain patch size, and the obtained subsequences are called patches. It is shown [17] that patching can effectively retain local semantic information and quadratically reduce computation and memory usage of the attention maps. However, such patching with fixed size has one fatal flaw: Once the input sequence is patched, the following TE can only learn the attention weights between patches of a fixed size, which fully did not exploit the relationships between multiscale patches at all, and may lead to a poor detection performance.

B. Main Contributions of Our Work

To address the above issues, different from the existing works, this article proposes a novel multiscale feature fusion TE-based ETD architecture MFFTD, based on semi-supervised learning paradigm consisting of self-supervised feature extraction and supervised fine-tuning. This approach leverages both unlabeled and labeled data, enhancing model performance even with limited labeled anomalous data. Especially, a novel multiscale feature fusion TE, named as hierarchical patching enhanced transformer encoder (HPTE), is designed to intentionally extract and fuse multiscale features. Since electricity consumption data typically contain patterns at different time scales, HPTE can effectively capture these complex temporal dependencies, thereby improving the identification of ET behaviors.

In detail, the main contributions of this article can be summarized as follows.

- 1) In MFFTD, the core component, the HPTE, is designed. In detail, at each HPTE layer, the input temporal sequence is patched with variable patch size, and then the multihead attentions are used to infer the correlation between any pair of patches. Then, hierarchically stacked HPTE layers are used to automatically

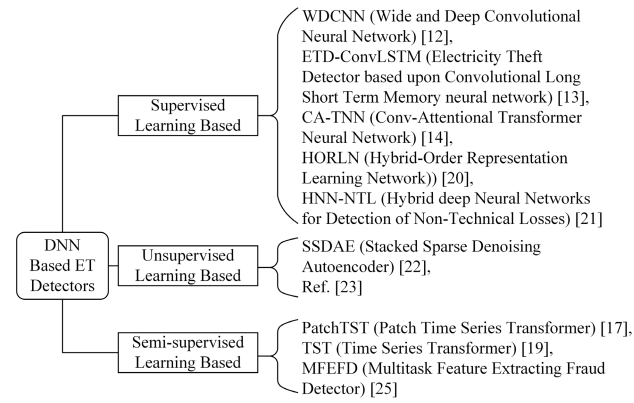


Fig. 1. Summary of DNN-based ET detectors categorized from learning paradigm.

extract multiscale features of the input long temporal sequence.

- 2) To accurately detect ET under the constraint of only limited number of labeled data available, MFFTD is trained in a semi-supervised manner. Specifically, a self-supervised pretrained scheme is implemented to learn the structure parameters of MFFTD, in which the self-supervised pretext task is intentionally designed to forecast the randomly masked segments in time series. Then, the penultimate of the pretrained model as the feature extractor (FE) and the added output layer as classifier are fine-tuned with a small percentage of labeled samples in a supervised way. In brief, this hybrid approach can merge the advantages of supervised and unsupervised learning paradigms and improve ETD performance and generalization.
- 3) Extensive experiments on multiple electricity consumption datasets are conducted to compare our proposed MFFTD scheme with various state-of-the-art (SOTA) methods. In terms of main performance metrics, the effectiveness of the proposed ETD scheme MFFTD is verified.

The rest of this article is organized as follows. In Section II, typical DNN (including transformer)-based ETD schemes are summarized and their disadvantages are discussed. Section III presents the proposed ETD framework MFFTD and describes its main modules in detail. Section IV provides the comprehensive performance comparison between the proposed MFFTD framework and SOTA ETD methods. Finally, we briefly conclude this article and point out future work.

II. RELATED WORK

DNN-based time-series anomaly detection schemes can automatically extract temporal features from historical electricity consumption data and then detect anomalies. Generally, from the perspective of learning paradigm, the existing schemes can be classified into three categories, as shown in Fig. 1: supervised-learning-based, unsupervised-learning-based, and semi-supervised-learning, which are correspondingly discussed in Sections II-A–II-C.

A. Supervised-Learning-Based Methods

Supervised-learning-based methods focus on building predictive models from labeled data to detect ET. There are

two mainly used DNNs: CNN and LSTM. Wide and deep CNNs (WDCNNs) [12] consist of a wide component and a deep component. The wide component extracts the global features of 1-D electricity consumption data, and the deep component captures the periodical and nonperiodical features from the transformed 2-D data matrix. Then the features from two components are added together to predict anomaly probability. But the wide component (i.e., the fully connected layer) cannot effectively learn temporal dependencies from customers' electricity consumption data [6], and the simple adding operation cannot effectively fuse two types of features. Hybrid-order representation learning network (HORLN) [20] learns the hybrid-order representations of the transformed 2-D data matrix. The first-order representations contain inter-and-intra week information, and the second-order representations are composed of global temporal dependencies learned from self-correlation matrices. Through integrating both sequential and nonsequential data, hybrid DNNs for detection of nontechnical losses (HNN-NTLs) [21] combines an LSTM network and a multilayer perceptrons (MLPs) to detect ET, in which LSTM analyzes the raw daily energy consumption history, MLP processes nonsequential data, and then the hybrid features are used to predict the anomaly probability. To capture both the global and local temporal dependencies of input time series, convolutional LSTM (ConvLSTM) is applied into ETD, in which convolutional operations are used into both the input-to-state and state-to-state transitions in LSTM [13].

Recently, since TE can automatically learn the relationships between elements in a long sequence through multihead attention mechanism, transformer has witnessed great application for processing sequential data, including ETD task. Conv-attentional transformer neural network (CA-TNN) [14] is a hybrid model of conv-attentional module and transformer model to extract local features and global features successively, and the final hybrid features are used for ETD. Specifically, the whole load sequence is divided into multiple segments, and the global features of consumption data are extracted by calculating the self-attention between segments. Moreover, conv-attentional module is used to embed the input data and capture the local features in each segment. The weakness of CA-TNN lies in that it can only capture dependencies of segments of a fixed size. In summary, although supervised paradigm is intentionally designed for anomaly detection using training dataset with explicit normal/anomalous labels, it may accurately conduct ETD. However, its significant weakpoint is that a large amount of labeled data are needed to train the detection model. Otherwise, the limited number of labeled data constrains the detection performance (overfitting to the small dataset). However, the availability of sufficient labeled temporal samples is extremely limited. Therefore, it is imperative to leverage the existing plethora of unlabeled data.

B. Unsupervised-Learning-Based Methods

For the category of unsupervised-learning-based ETD, stacked sparse denoising autoencoder (SSDAE) [22] uses a stacked sparse denoising autoencoder to learn feature of normality through minimizing reconstruction error and takes the testing samples with the error higher than the threshold as anomalies. Similarly, [23] incorporates LSTM into autoencoders and is also based on reconstruction method.

Unsupervised-learning-based methods usually aim to learn the latent distribution of unlabeled data. The objective function of the data reconstruction is designed for dimension reduction or data compression, rather than anomaly detection.

As a result, purely unsupervised paradigm did not assume the existence of anomalous temporal samples. Its main goal is not specially designed for anomaly detection, but for generally learning the normality of time series [24], which may lead to relatively poor ETD performance.

C. Semi-Supervised-Learning-Based Methods

To integrate the benefits of supervised and unsupervised paradigms under the constraint of insufficient labeled data, semi-supervised-learning-based methods are proposed, which fully exploit both labeled and unlabeled data to build the predictive models. Multitask feature extracting fraud detector (MFEFD) [25] simply uses the stacked fully connected layers as DNN structure and is trained in a semi-supervised manner, which combines supervised and unsupervised learning through multitask learning. Recently, some unified TE-based semi-supervised learning frameworks are proposed [17], [19], which can be naturally used in ETD task. Specifically, TST [19] presents an unsupervised pretraining framework for multivariate time series, in which segments of different sizes are masked and the model is trained to predict the masked contents with mean square error loss, and offers substantial performance benefits over fully supervised learning on downstream tasks.

However, the original TE architecture used in TST can only learn temporal dependencies between each data point at different timestamps, which neglects the local semantic information. Besides, the efficiency of the method is limited by the length of the time series (L) with computational complexity overhead as $O(L^2)$. To solve the above issue, before feeding the temporal sequence to TE, the latest work PatchTST [17] segments the long time series into subseries-level patches as input tokens to transformer. The so-called patching design can both retain the local semantic information in embedding and quadratically reduce computation and memory usage. However, the total number of patches is prefixed, which can only capture the relationships between patches in a fixed granularity way. Different from PatchTST, our work designs an HPTE with variable patch sizes in each HPTE layer, which can effectively extract multiscale latent temporal dependencies in time series and greatly improve the ETD performance. Furthermore, the impact of variable patch size in each HPTE layer is investigated.

III. MFFTD: PROPOSED METHOD FOR ETD

A. Overall Architecture

Fig. 2 presents the overview framework of the proposed MFFTD scheme, which consists of training phase and operating phase: the purpose of training phase is to train the MFFTD framework, and then the operating phase uses the trained MFFTD to conduct ETD. The training phase is explicitly composed of two modules: self-supervised pretraining and supervised two-stage fine-tuning. In module 1, a self-supervised pretraining regression task is used to learn the structural parameters of HPTE used as the FE. In module 2, the pretrained FE in module 1 and classifier are fine-tuned in a supervised-based two-stage way: in the first stage, the

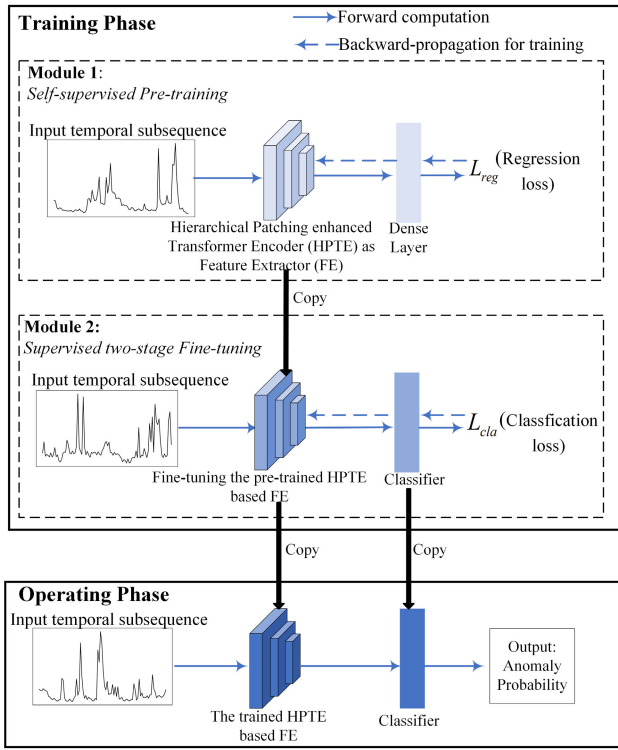


Fig. 2. Overall framework of the proposed MFFTD with training and operating phases.

parameters of pretrained FE are frozen, and only the classifier is trained; and in the second stage, both are trained, which can protect the pretrained parameters of FE and stabilize the whole training processes. In the operating phase, the trained FE and classifier, i.e., our scheme MFFTD takes load profile as input and outputs its anomaly probability. Electricity customers with anomaly probability larger than 0.5 are judged as illegal.

Note that the designed HPTE is the core component in our framework shown as Fig. 2, which is specifically described in Fig. 3.

B. Hierarchical Patching Enhanced Transformer Encoder

In Fig. 2, HPTE is used as the FE, which is composed of multiple stacked HPTE layers. The i th HPTE layer is shown in Fig. 3, which consists of two modules: hierarchical patch embedding (HPE) and TE. In the i th HPTE layer, the length of input sequence is denoted as L_{i-1} , the length of output sequence as L_i , the embedding dimension as d , and the patch size as p_i . The first (i.e., gray) rectangle represents the input feature matrix F_{i-1} of the i th layer, the second (i.e., blue) rectangle represents the feature matrix F_i^{HPE} output by HPE, and the third (i.e., red) rectangle represents the final output feature matrix F_i of the i th layer. Note that $L_i = (L_{i-1}/p_i)$. The detailed operations of HPE and TE modules are as follows.

1) *Hierarchical Patch Embedding*: Given the input feature of the i th HPTE layer, $F_{i-1} \in \mathbb{R}^{L_{i-1} \times d}$, HPE module, using patch size p_i , first divides the input feature into multiple, i.e., (L_{i-1}/p_i) patches, each patch $P \in \mathbb{R}^{p_i \times d}$. Then all the patches are put together and reshaped into a matrix $P' \in \mathbb{R}^{(L_{i-1}/p_i) \times (p_i \times d)}$, which are further projected into a feature map F_i^{HPE} with length (L_{i-1}/p_i) formulated in HPE module shown

in Fig. 3, that is, $F_i^{\text{HPE}} = P'W + b$, where W and b are learnable parameters.

Note that it is shown that transformer with positional encoding can hardly preserve temporal information [26], and moreover, we empirically found out that adding both learnable and fixed positional encoding each layer actually reduces, instead of improving, detection performance. Our empirical results comply with the conclusion obtained by [26] that the vanilla transformer is not sufficiently effective for modeling time series, which acts as the rationale that our work patches time series with multiple variable patch sizes, to effectively capture the multiple-scale temporal relationships in time series.

2) *Transformer Encoder*: In TE module, the existing TE [27] is used to characterize the relationships between different patches, which, as shown in Fig. 3, consists of a multihead attention (MHA) layer and an MLP layer. Specifically, each head in the MHA transforms the feature map F_i^{HPE} into query matrices Q_h , key matrices K_h , and value matrices V_h . The scaled dot-product attention [Attention(.)] is conducted for each head and outputs the concatenated matrices [MHA(.)] as follows:

$$\begin{aligned} \text{Attention}(Q_h, K_h, V_h) \\ = \text{Softmax}\left(\frac{Q_h K_h^T}{\sqrt{d_k}}\right) V_h \end{aligned} \quad (1)$$

$$\begin{aligned} \text{MHA}(F_i^{\text{HPE}}) \\ = \text{Concat}(\text{head}_1, \dots, \text{head}_H) W_O + b_O, \quad \text{where} \\ \text{head}_h \\ = \text{Attention}\left(F_i^{\text{HPE}} W_h^Q, F_i^{\text{HPE}} W_h^K, F_i^{\text{HPE}} W_h^V\right) \end{aligned} \quad (2)$$

where Softmax is the softmax activation function, Concat is the concatenation operation, d_k is the embedding dimension of K_h , and H is the number of the heads. W_O , W_h^Q , W_h^K , W_h^V , and b_O are learnable parameters.

Besides, the residual connecting is used to avoid gradient explosion as follows:

$$V' = \text{Layernorm}(F_i^{\text{HPE}} + \text{MHA}(F_i^{\text{HPE}})) \quad (3)$$

$$F_i = \text{Layernorm}(V' + \text{MLP}(V')) \quad (4)$$

where Layernorm refers to layer normalization, a technique used to normalize the inputs to a layer in a neural network. F_i is the output of i th HPTE layer, and V' is an intermediate variable.

C. Training of MFFTD: Self-Supervised Pretraining and Supervised Fine-Tuning

Specifically, we consider the load profile as an univariate sequence of T data points $X = \{x_1, \dots, x_T\}$. When detecting whether x_t is an anomaly, i.e., $y_t \in \{0, 1\}$ (1 denotes an anomalous data point), we take into account the dependencies within a local contextual window of length L , i.e., $w_t = \{x_{t-L+1}, \dots, x_t\}$ to help the model learn to detect anomalies. Here we simply drop the window w_t when $t < L$ [28].

1) *Self-Supervised Pretraining as Regression Task*: To strengthen the FE's ability to capture underlying features of unlabeled data, the regressive task of recovering the masked input is used: the mathematical essence is using the relationships between patches to recover the missing contents.

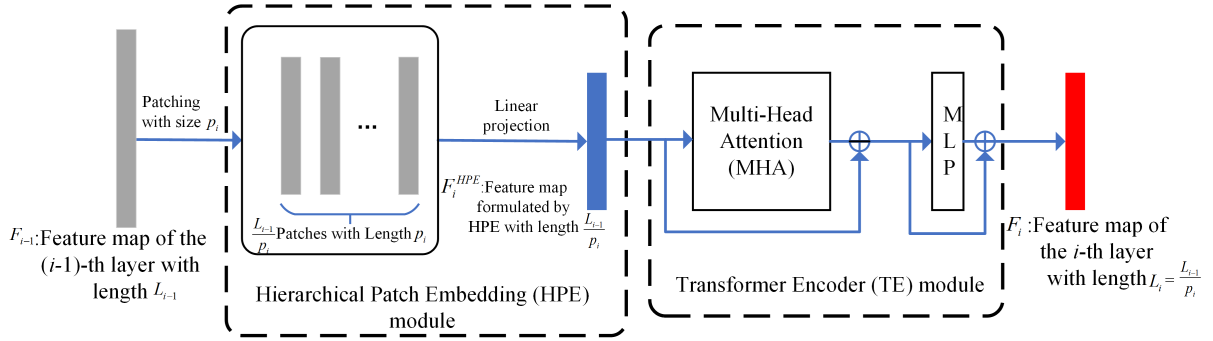


Fig. 3. Illustration of the structure of the i -th HPTE layer and its processing.

Given the detection window $w_t \in \mathbb{R}^{L \times 1}$, the masked input $\hat{w}_t$ for FE can be obtained as follows:

$$\hat{w}_t = M \odot w_t \quad (5)$$

where $\odot$ is the Hadamard product, and $M \in \mathbb{R}^{L \times 1}$ is a binary mask which consists of alternating segments of 0 s and 1 s. The length of segments of 0 s follows geometric distribution of mean l_0 : when the segment of 0 s stops, it is succeeded by a segment of 1 s, whose length follows geometric distribution of mean l_1 . Let r denote the proportion of the mask to be 0, thus $l_1 = ((1-r)/r)l_0$.

The masked detection window $\hat{w}_t$ is then fed to the FE. Afterward, the extracted feature F is fed to an FC layer and flattened to output its forecasting of the partially masked input window $\hat{w}_t$, denoted as $\tilde{w}_t$ given as (6). The predictive error on the masked values (with indices in the set $M_0 \equiv \{i : M(i) = 0\}$) is regarded as regression loss (i.e., mean square error loss), shown as (7)

$$\tilde{w}_t = \text{Flatten}(FW + b) \quad (6)$$

$$L_{\text{reg}} = \sum_{i \in M_0} [\tilde{w}_t(i) - w_t(i)]^2 \quad (7)$$

where Flatten is the flatten operation, and W and b are learnable parameters.

Note that the objective of the self-supervised pretraining is to bootstrap the structural parameters of FE for the successive supervised fine-tuning. That is, the FC layer used for outputting the masked temporal points will be abandoned in the following supervised fine-tuning.

2) *Two-Stage-Based Supervised Fine-Tuning*: The self-supervised training learns the structure parameters of the FE but leaves the classifier untrained after being initialized. To better fit the ETD task, the whole MFFTD framework is fine-tuned using the small percentage of the labeled data. However, the different task goal, i.e., regressive task in pretraining and anomaly detection (classification) downstream task, makes the whole model hard to converge. Thus, we apply a two-stage training procedure to train the classifier (implemented as a linear layer and a sigmoid activation function) and the FE after self-supervised pretraining as shown in Algorithm 1. The two-stage training can avoid the condition that large gradient updates triggered by the randomly initialized classifier will wreck the learned parameters in FE [25]. Specifically, in the first stage (lines 3–10 in Algorithm 1), we freeze the parameters of FE obtained

Algorithm 1 Two-Stage-Based Supervised Fine-Tuning

Input: (W, Y) the labeled time windows, ME_1 the maximal epochs of training Classifier only, ME_2 the maximal epochs of training both FE and Classifier, θ_{FE} the pre-trained parameters of FE

Output: $\theta = (\theta_{FE}, \theta_c)$ parameters of the whole MFFTD

- 1: Initialize FE and classifier with parameters θ_{FE}, θ_c , respectively
 - 2: Initialize count variables $c_1 \leftarrow 0, c_2 \leftarrow 0$
 - 3: **Stage 1: Train the Classifier (Freezing the FE)**
 - 4: **for** $c_1 = 1$ to ME_1 **do**
 - 5: **for** each batch (w_t, y_t) in (W, Y) **do**
 - 6: The values of
 - 7: p^1, L_{cla}^1 are presented in (8) and (9)
 - 8: $\theta_c \leftarrow \text{Make a gradient step to minimize } L_{cla}^1$
 - 9: **end for**
 - 10: **end for**
 - 11: **Stage 2: Fine-Tune the Entire Model (FE+Classifier)**
 - 12: **for** $c_2 = 1$ to ME_2 **do**
 - 13: **for** each batch (w_t, y_t) in (W, Y) **do**
 - 14: The values of
 - 15: p^2, L_{cla}^2 are presented in (8) and (9)
 - 16: $\theta_{FE}, \theta_c \leftarrow \text{Make a gradient step to minimize } L_{cla}^2$
 - 17: **end for**
 - 18: **end for**
 - 19: **return** $\theta = (\theta_{FE}, \theta_c)$
-

by pretraining and only train the classifier; in the second stage (lines 11–19 in Algorithm 1), we train the whole MFFTD, including the parameters of FE and classifier). The two-stage training shares the same model architecture (FE as shown in Fig. 3) and loss function as (9). Given the input window w_t and its label y_t , FE outputs the extracted features F , and classifier takes as input the flattened features, and outputs the anomaly probability p (when p is more than 0.5, the input window w_t is judged as an anomaly) as follows:

$$p = \sigma(\text{Flatten}(F)W + b) \quad (8)$$

where σ is the sigmoid activation function, Flatten is the flatten operation, and W and b are learnable parameters. Considering the data imbalance between normal and anomalous temporal data, the weighted binary cross-entropy is adopted to evaluate the difference between the predicted probability and true label

as follows:

$$L_{\text{cla}} = -\frac{1}{n} \sum_{i=1}^n w_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)] \quad (9)$$

where n is the batch size, and $w_i \in \{0.1, 0.9\}$ is the weight with 0.9 for an anomaly. y_i is the true label, and p_i is the predicted probability.

IV. EVALUATION EXPERIMENTS

A. Experimental Settings

In our work, a Python-based software platform is developed to simulate multiple and typical ET behaviors in real-world electricity consumption data, based on which our proposed MFFTD scheme and the SOTA ETD baselines are implemented and comprehensively evaluated. In detail, this section first describes the publicly used datasets, including the IRISH dataset (given in Section IV-A) and the State Grid Corporation of China (SGCC) dataset (given in Section IV-E). Then, six false data injection (FDI) attacks that simulate multiple typical ET behaviors are clearly provided in Section IV-A2. The commonly used evaluation metrics are given in Section IV-A3. Finally, the implementation details are listed in Section IV-A4, including hardware and software configurations and hyperparameters' settings.

1) *Smart Meter Datasets*: The smart meter dataset adopted for our performance evaluation is publicly available from the Irish CER smart metering project [29] which is widely used in the research of ETD. The dataset contains more than 500 days of smart meter data collected from over 5000 residential users and small- and medium-sized business users during 2009 and 2010. The dataset is considered as not tampered and can be used as ground truth because all the users completed the pretrial or posttrial surveys. The sampling rate of the smart meter is one temporal data point every half an hour such that a daily load profile consists of 48 data points. In the experiments, we use the collected data of 365 days, drop the data of any user with nan (nan represents a null value), recover the incorrect data, and normalize the data with min-max normalization [30]. The detection window is set to seven days, i.e., $L = 336$ instead of one day to improve the performance of the detection. We divide 5000 users into 4000, 500, and 500 users for the pretraining set, validation set, and test set. In each set, the whole electricity consumption data are viewed as a long time series. In the pretraining set, we randomly pick 1% daily electricity consumption data tampered with certain attacks, and pick 9% as honest daily electricity consumption data. In summary, the proportion of unlabeled, fraudulent, and honest samples are 90%, 1%, and 9%, respectively. The pretraining set is used to learn the structure parameters of FE, and labeled samples in the fine-tuning set are used to fine-tune the pretrained model. In the validation set and test set, we randomly pick 10% daily electricity consumption data tampered with certain attacks. The left ones are normal data. The validation set is used to judge whether overfitting happens and choose the best learned parameters. The test set is used to evaluate performance.

2) *Attack Model*: The tampering behaviors of ET on load profiles can be simulated by various types of FDI attacks. In this study, we consider six types of FDI attacks widely used

TABLE I
SIX TYPES OF FDI ATTACKS

Attack Type	Modification
1	$\tilde{x}_t = \alpha_t x_t, \alpha_t = \text{random}(0.2, 0.8)$
2	$\tilde{x}_t = \max(x_t - \gamma, 0), \gamma < \max(X)$
3	$\tilde{x}_t = \min(x_t, \gamma), \gamma < \max(X)$
4	$\tilde{x}_t = \alpha_t x_t, \alpha_t = \begin{cases} 0, & t_{\text{start}} < t < t_{\text{start}} + \Delta t \\ 1, & \text{otherwise} \end{cases}$ $\Delta t = \text{random}(0.2, 0.8) \cdot n,$ $t_{\text{start}} = \text{random}(0, n - \Delta t)$
5	$\tilde{x}_t = \alpha \text{mean}(X), \alpha = \text{random}(0.2, 0.8)$
6	$\tilde{x}_t = x_{n-t}$

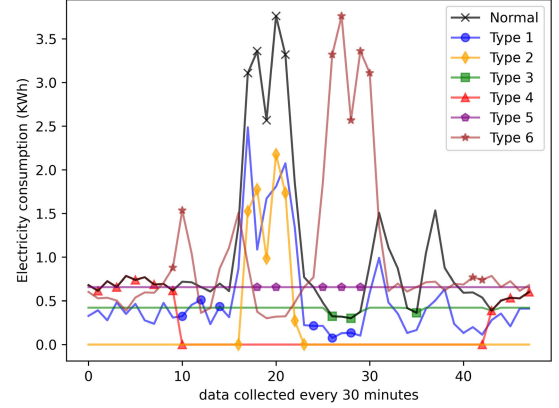


Fig. 4. Example of daily electricity consumption and its tampered data corresponding to six types of FDI attacks.

in [14] and [25] which are defined in Table I. In Table I, $\tilde{x}_t$ and x_t are tampered and real electricity consumption at timestamp t , respectively. $X = \{x_1, x_2, \dots, x_{48}\}$ is the daily electricity consumption of 48 data points collected by the smart meter. $\max(X)$, $\min(X)$, and $\text{mean}(X)$ are the maximum value, minimum value, and average value of the daily electricity consumption X , respectively.

Basically, there are two types of consumption attacks, called reduced consumption attacks and load profile shifting attacks [6]. Reduced consumption attacks (types 1–4) directly reduce the electricity consumption data, while load profile shifting attacks (types 5 and 6) shift the consumption data at a high electricity price to the consumption data at a low electricity price. As shown in Table I, type 1 attack multiplies the electricity consumption at the timestamp t with a changing-with- t parameter α_t ranging from 0.2 to 0.8. Type 2 attack reduces the daily electricity consumption by a randomly chosen value γ smaller than the maximum value of the daily consumption and keeps the positive values. Type 3 attack retains the value no bigger than γ . Type 4 attack picks a time period and sets the data over the time period to zero. Type 5 attack replaces the raw data with the average value of the daily consumption multiplied with a parameter $\alpha \in (0.2, 0.8)$. Type 6 attack inverts the whole daily consumption. An example of the daily electricity consumption and its tampered data corresponding to six types of FDI attacks are shown in Fig. 4.

3) *Evaluation Metrics*: We adopt the F1 score, the area under a receiver operating characteristic (ROC) curve (AUC), recall, and false positive rate (FPR) as evaluation metrics. The detection results can be divided into four categories, including

TABLE II
PERFORMANCE OF DIFFERENT METHODS

Methods	Metrics					
	F1	AUC	Recall	FPR	Training time(s)	Testing time(s)
WDCNN	0.641	0.922	0.748	0.072	166	3.1
TST	0.723	0.945	0.745	0.040	890.1	3.6
PatchTST	0.779	0.950	0.840	0.045	99.8	1
CA-TNN	0.809	0.963	0.812	0.023	589.8	2.4
MFFTD	0.848	0.972	0.862	0.019	391.5	1.4

The bold results indicate the best performance.

true positive (TP), false positive (FP), false negative (FN), and true negative (TN). Here, we take predicting illegal users as positive, so that the meanings of TP, FP, FN, and TN are as follows:

- 1) *TP*: Number of illegal users correctly predicted as illegal users.
- 2) *FP*: Number of benign users wrongly predicted as illegal users.
- 3) *FN*: Number of illegal users wrongly predicted as benign users.
- 4) *TN*: Number of benign users correctly predicted as benign users.

Based on the above concept, precision, recall, and FPR are defined to reflect the model's ability to make the true detection of the ET behavior, where Precision = (TP/(TP + FP)), Recall = (TP/(TP + FN)), and FPR = (FP/(FP + TN)). A high precision can significantly reduce the additional costs incurred due to error detection and subsequent manual inspections; a high recall means that most of the anomalies are detected; and a good model should have a high recall and a low FPR. Then the F1 score, which is a comprehensive consideration of both precision and recall, is defined as follows:

$$\text{F1 score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \quad (10)$$

AUC represents the area under the ROC curve, which plots the TP rate against the FPR at various threshold settings. A higher AUC indicates better discrimination ability of the model, with an AUC of 1 representing a perfect classifier.

4) *Implementation Details*: We implement all the methods with Python-3.8.17, PyTorch-2.0.1, and PyTorch-Ignite-0.3.0. We use the Adam optimizer to train all the models with a learning rate of 0.001. We implement all the methods three times on an Intel i3-12100F CPU and an NVIDIA GeForce RTX 2060 12G GPU and report the results as the mean of six FDI attacks. We use the hyperparameters of the baseline models as presented in their respective papers. We choose the following hyperparameters for our method.

- 1) Dimension of the embedding $d = 128$.
- 2) Number of the heads $H = 8$.
- 3) Number of the layers $N = 3$.

Note that the rationale of selecting these hyperparameters is given in Section IV-C.

B. Performance Comparison

In this part, we compare the proposed MFFTD with four SOTA ET methods, of which two are supervised learning methods and two are semi-supervised methods. We implement all these methods on the preprocessed data for fair comparison.

1) *Time-Series Transformer*: TST [19] uses the TE and computes the relationship between data points at different timestamps. TST is trained in the semi-supervised way. In the unsupervised pretraining, TST completes the autoregressive task of forecasting the masked temporal data points and is implemented with $l_m = 8, r = 0.3$. In supervised fine-tuning, TST is trained to predict the probability of input being an anomaly.

2) *Patch TST*: PatchTST [17] takes time-series segments ($p = 8$) as input and is similarly trained in a semi-supervised way. In the unsupervised pretraining, the model learns to predict the randomly removed patches. In the supervised fine-tuning, the operation is the same as TST.

3) *Conv-Attentional Transformer Neural Network*: CA-TNN [14] divides the time series into several segments ($p = 8$), uses conv-attentional module to embed the segments, and uses transformer to extract global features by calculating the self-attention between each segment. It is trained in a supervised way to predict the anomaly probability of input.

4) *Wide and Deep CNN (WDCNN)*: WDCNN [12] consists of a wide component (a fully connected layer) and a deep CNN component (stacked convolutional layers). We implement the model with the length of fully connected layer in the wide and deep CNN components as 90 and 90, respectively, the number of filters in the convolutional layer as 60, and the number of convolutional layers as 5.

Table II provide the performance comparison results of different methods. The following results can be observed.

- 1) On average, MFFTD achieves the best results (0.848 in F1 score, 0.972 in AUC, 0.862 in recall, and 0.019 in FPR), with improvement of 3.8% in F1 score, 1.3% in AUC, 2.6% in recall, and 17.3% in FPR over the second-best method CA-TNN (0.809 in F1 score, 0.963 in AUC, 0.812 in recall, and 0.023 in FPR).
- 2) CA-TNN uses a conv-attentional module to automatically extract and implicitly fuse different types of features in electricity consumption data ahead of TE [14], which shares a similar idea with MFFTD (adopting HPTE to fuse multiscale features). The advantage of our proposed MFFTD is to explicitly extract and fuse multiple-scale features with variable patch size, and therefore, MFFTD averagely outperforms CA-TNN.
- 3) Compared with TST, PatchTST roughly extracts the temporal relationships in an ordered time-series through one-shot and fix-sized patching, thus PatchTST performs better than TST.

The number of parameters, average training time, and average testing time for all the methods are also empirically given in Table II for comparison. The number of trainable

TABLE III
THEORETICAL ANALYSIS OF PER-LAYER COMPLEXITY
FOR DIFFERENT METHODS

Method	Complexity per Layer
WDCNN	$O(k \times l \times d^2)$
TST	$O(l^2 \times d)$
PatchTST	$O((\frac{l}{p})^2 \times d)$
CA-TNN	$O(k \times l \times d^2) + O((\frac{l}{p})^2 \times d)$
MFFTD	$O((l^*)^2 \times d)$

l is the constant sequence length, l^* is the variable sequence length, p is the patch size, k is the kernel size of CNN and d is the embedding dimension.

parameters is derived from the neural network architectures, encompassing both the FE and classifier components. Training time reflects the duration required for the complete training process across all the samples over 100 epochs on a GPU, while testing time represents the time needed for a single prediction for all the samples.

The theoretical analysis of per-layer complexity for different methods is given in Table III. In our study, the sequence length l is much bigger than embedding dimension d , thus theoretically speaking, the training time of WDCNN should be smaller than TST, which is consistent with our experimental results. CA-TNN is a hybrid model consisting of CNN and TE with fixed patching size, such that its training time is larger than WDCNN while being less than TST. For our proposed MFFTD, the computational complexity changes to $O(\sum_{i=1}^N l_i^2 \times d)$, where l_i is the sequence length for the i th layer and different patching modes can greatly influence the computational complexity, which is discussed in Section IV-D.

C. Analysis of the Selection of Main Hyperparameters

We conduct extensive experiments on the impact of three important hyperparameters in MFFTD to empirically seek the best settings. The results are presented in Table IV. As shown in Table IV, the range of values for dimension of the embedding d and number of heads H are the same as TST [19] and the best results are almost consistent, i.e., $d = 128$, $H = 8$. The number of the HPTE layers decides the scales of fused features. Empirically, moderate scales are suitable for anomaly detection. The reason may lie in that our scheme uses the variable patch size in each HPTE layer, and therefore, the larger number of layers may lead to shorter size of each patch, which results in TE focusing on local feature correlations, vice versa for the smaller number of layers. In brief, empirically, the dimension of the embedding, number of the heads, layers, and patch size for each layer are chosen as 128, 8, 3, and $p_1, p_2, p_3 = 2, 3, 8$, respectively, in our experiments.

D. Ablation Analysis

An ablation study is conducted to demonstrate the impact of the main components in the proposed MFFTD framework on the ETD performance. Specifically, two variants of our MFFTD are implemented as shown in Table V.

1) *Variant MFFTD-H: MFFTD Without HPE*: MFFTD-H directly uses TE instead of HPTE and is trained with our proposed semi-supervised way.

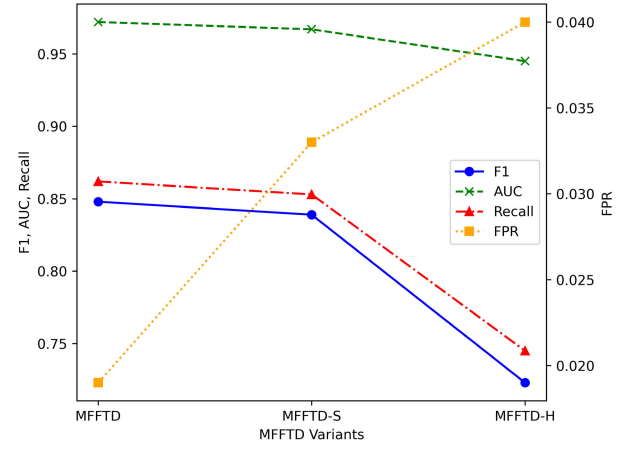
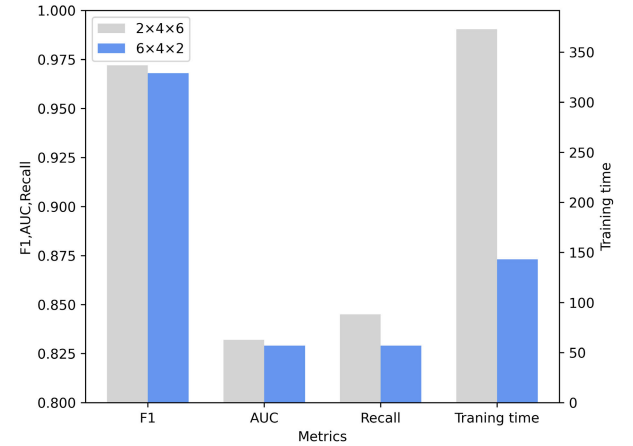
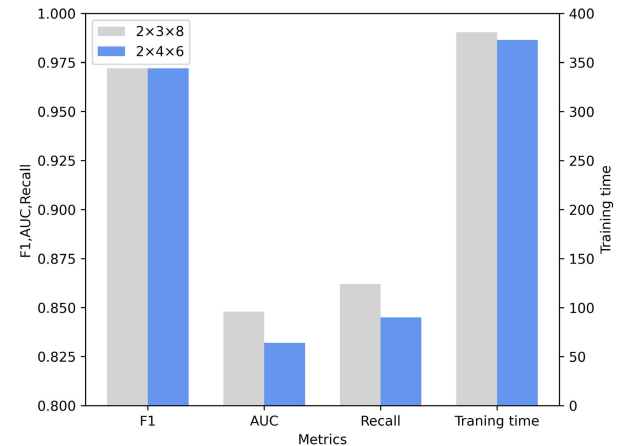


Fig. 5. Ablation analysis of MFFTD.



(a)



(b)

Fig. 6. Specific contribution to model performance in different settings of multiscale patching. (a) Comparison between two patching modes: increasing mode and decreasing mode. (b) Comparison within increasing mode.

2) *Variant MFFTD-S: MFFTD Without Self-Supervised Pre-training*: MFFTD-S keeps the neural network architecture, but is directly trained in a supervised way.

TABLE IV
PERFORMANCE OF THREE PRIMARY HYPERPARAMETERS

Hyperparameter	Setting	Metrics					
		F1	AUC	Recall	FPR	Training time(s)	Testing time(s)
Dimension of the embedding	64	0.850	0.970	0.856	0.018	263.1	1.4
	128	0.848	0.972	0.862	0.019	391.5	1.4
Number of the heads	8	0.848	0.972	0.862	0.019	391.5	1.4
	16	0.836	0.972	0.858	0.013	408	1.4
Number of the layers	2	0.813	0.964	0.825	0.023	91.8	1
	3	0.848	0.972	0.862	0.019	391.5	1.4
	4	0.837	0.972	0.859	0.022	490.6	1.5

The bold results indicate the best performance.

TABLE V
STRUCTURE OF MFFTD VARIANTS

Components	HP	Self-supervised Pre-training
Variants		
MFFTD	✓	✓
MFFTD-H	×	✓
MFFTD-S	✓	×

Fig. 5 illustrates the average F1 score, AUC, recall, and FPR of six ET attacks and the following results can be obtained.

- 1) MFFTD-H performs much worse, compared with MFFTD, with nearly 14.7% drop in terms of F1 score, 2.7% drop in terms of AUC, 13.5% drop in terms of recall, and 110.5% growth in terms of FPR. It demonstrates that the designed HPTE can effectively fuse multiscale feature thus improving the detection performance.
- 2) MFFTD achieves better performance than MFFTD-S, which implies that semi-supervised learning effectively leverages the underlying distribution and general patterns of both labeled and unlabeled data to enhance predictive performance over a broader dataset. The ablation results show that our proposed semi-supervised training strategy can improve the ETD performance.

We also provide the results of two patching modes as shown in Fig. 6 to analyze each specific contribution to model performance. In detail, the temporal sequence of length 48 (i.e., the number of temporal data points gathered in a whole day) is hierarchically divided into three layers, i.e., $p_1 \times p_2 \times p_3 = 48$, and two kind of patching strategies are designed: decreasing mode $p_1 \geq p_2 \geq p_3$ and increasing mode $p_1 \leq p_2 \leq p_3$.

The following results can be empirically obtained.

- 1) Increasing mode is better than decreasing mode in terms of performance metrics: Fig. 6(a) shows that the setting $p_1 \times p_2 \times p_3 = 2 \times 4 \times 6$ achieves 1.9% higher recall than the opposite setting. However the computational complexity is $O(\sum_{i=1}^N l_i^2 \times d)$. In the increasing mode, the sequence length l_i for the i th layer is normally bigger than that in decreasing mode. So it takes more time to train the whole model.
- 2) For the detailed choice in increasing mode, we empirically find from Fig. 6(b) that the model performs best under the setting where the front patching size is slightly smaller, and the rear patching size increases significantly. The observations are analogous to the

TABLE VI
PERFORMANCE OF DIFFERENT METHODS ON THE SGCC DATASET

Methods	Metrics			
	F1	AUC	Recall	FPR
WDCNN	0.592	0.824	0.727	0.139
TST	0.623	0.845	0.705	0.140
PatchTST	0.646	0.854	0.703	0.086
CA-TNN	0.588	0.825	0.716	0.137
MFFTD	0.651	0.859	0.708	0.085

The bold results indicate the best performance.

design of the well-known architecture of ResNet [31], in which the receptive field is also in an increasing way. The rationale of beginning with small patch size lies in that in early layers, the local features should be extracted through small patch size, and as the network deepens, more extract feature should be emphasized through large patch size.

E. Performance Evaluation on the SGCC Dataset

We further evaluated the performance of MFFTD alongside the same baseline methods on another dataset: the SGCC [12]. This dataset contains the electricity consumption data of 42372 electricity customers within 1035 days (from January 1, 2014 to October 31, 2016) released by SGCC (<http://www.sgcc.com.cn/>). We use the same data preprocessing methods for the Irish dataset, and for details refer to Section IV-A.

Performance results are shown in Table VI. Similarly, our proposed MFFTD achieves the best performance among all the schemes, which is consistent with the results obtained on the Irish dataset.

V. CONCLUSION AND FUTURE WORK

This article proposes a novel semi-supervised-learning-based multiscale feature fusion TE for effectively and efficiently detecting ET, MFFTD. Specifically, the designed hierarchical patching with variable patch size can fuse the multiscale feature of the time series, and thus greatly facilitate the effective ETD. Furthermore, MFFTD is trained in a semi-supervised manner, i.e., self-supervised pretraining and supervised fine-tuning, which can efficiently leverage both abundant unlabeled temporal data and a small percentage of labeled data available. Specifically, unlabeled data are used in the pretraining process to extract general feature to assist the downstream ETD task. Experiments are conducted on a real-world electricity consumption dataset,

and the results show that our proposed ETD framework MFFTD can effectively detect the anomalous electricity behaviors and outperforms SOTA other transformer-based ETD methods.

The future work includes the following aspects. First, our work uses the historical load profile to train the ETD model, which is essentially univariate time-series anomaly detection. If other features would be available, e.g., customers' traits and weather information, multivariate and their relationships can be explored for more accurate ETD. Second, our work follows the semi-supervised learning paradigm: Pretraining with pretext task, and fine-tune using small percentage of labeled anomalous samples from downstream task, it can be effective for known anomaly types, but may be vulnerable to the emerging unknown anomaly types. One potential solution is to learn data representations assisted by a few labeled anomalous samples: The representations are optimized such that these normal instances are located in dense neighborhoods around the designated origin in latent space, while the anomalous samples are distributed far from origin. Then, the distance of the representation of sample to the origin is viewed as its anomaly score for ETD.

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